

A Permutation-Group Counterexample to Interpolation of Cocompactness

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Status. Candidate full counterexample, likely valid, requiring human review. The construction gives a negative answer to the arbitrary-group interpolation question in source Remark 1.7, for both the complex method and the real method.

Abstract

Cwikel and Tintarev ask whether cocompactness persists under interpolation for every choice of the acting group, without the additional mollifier hypotheses in their theorem. It does not. Take both Banach couples to be (ℓ_∞, ℓ_1) and let the group consist of all coordinate permutations. The endpoint identity on ℓ_1 is cocompact because ℓ_1 has the Schur property. For every $0 < \theta < 1$, however, both complex interpolation and the real method with fine index $1/\theta$ yield $\ell_{1/\theta}$. Normalized indicators of growing finite sets have norm one but are weakly null after every sequence of coordinate permutations. Thus the interpolated identity is not cocompact.

1 Source question

The source is Cwikel and Tintarev’s *On interpolation of cocompact imbeddings* [1]. We use arXiv:1008.4371v3. For a group $\mathcal{D}$ of continuous linear bijections of a Banach space A , the paper defines

$$u_k \xrightarrow{\mathcal{D}} u \iff g_k(u_k - u) \rightarrow 0 \text{ for every sequence } (g_k) \subset \mathcal{D}. \quad (1)$$

An embedding $A \hookrightarrow B$ is $\mathcal{D}$ -cocompact when every sequence converging in the sense of (1) converges in the norm of B .

Theorem 2.3 proves persistence of cocompactness under real and complex interpolation assuming, among other baseline hypotheses, the existence of a $\mathcal{D}$ -covariant mollifier family. Remark 1.7 asks whether the additional conditions can be removed so that persistence holds for every group $\mathcal{D}$, and explicitly anticipates that the answer may be negative.

2 Proof intuition

The two endpoints play opposite roles. At the strong endpoint, the Schur property of ℓ_1 turns ordinary weak convergence into norm convergence. Because the identity belongs to every group, this immediately makes the identity on ℓ_1 cocompact relative to any group of isometries.

Interpolation with ℓ_∞ destroys the Schur property. In the intermediate ℓ_p space, distribute unit mass evenly over k coordinates. The resulting vector still has norm one. A permutation may move its support to any k -element set, but it cannot concentrate the individual coefficient, which is only $k^{-1/p}$. Splitting any dual vector into a finite head and a small tail proves weak convergence uniformly over all such moved supports.

The main result of this paper, Theorem 2.3, deals with persistence of cocompactness for interpolated spaces. It can be considered as a sort of counterpart to results about persistence of compactness for operators mapping between “real method” or “complex method” interpolation spaces, in particular those in Section 9.6 of [8] and in [29], in which hypotheses having a partial analogy with hypotheses of Theorem 2.3 are imposed.

Remark 1.7. Note however that the compactness results of [8] and [29] were subsequently found to also hold without these kinds of hypotheses and/or under other alternative hypotheses. (See e.g. [9, 10, 13, 12, 11] and the references therein.) An analogous complete removal of additional conditions in the case of cocompactness would mean that persistence of cocompactness under interpolation holds for all choices of the group $\mathcal{D}$, which remains an open question. A negative answer to it would not surprise us.

As examples of applications of Theorem 2.3, we prove the cocompactness of classical Peetre imbeddings [26] of inhomogeneous Sobolev spaces with fractional

Figure 1: The open question in arXiv:1008.4371v3, PDF page 3.

3 The construction

For a permutation π of $\mathbb{N}$, let

$$(P_\pi x)_j = x_{\pi^{-1}(j)}, \quad \mathcal{D} = \{P_\pi : \pi \in \text{Sym}(\mathbb{N})\}.$$

Every P_π is a linear bijective isometry on ℓ_r for $1 \leq r \leq \infty$. Set

$$(A_0, A_1) = (B_0, B_1) = (\ell_\infty, \ell_1). \tag{2}$$

Thus $A_1 \hookrightarrow A_0$, all endpoint inclusions $A_j \hookrightarrow B_j$ are identities, and the same group acts isometrically on all four endpoints.

Lemma 1. *The identity embedding $\ell_1 \hookrightarrow \ell_1$ is $\mathcal{D}$ -cocompact.*

Proof. Suppose $x_k \xrightarrow{\mathcal{D}} 0$ in ℓ_1 . Since the identity operator belongs to $\mathcal{D}$, definition (1) implies $x_k \rightharpoonup 0$ ordinarily. The Schur property of ℓ_1 says that weakly convergent sequences converge in norm [3]. Hence $\|x_k\|_1 \rightarrow 0$, exactly as required. $\square$

Lemma 2. *Let $1 < p < \infty$ and define*

$$u_k = k^{-1/p} \sum_{j=1}^k e_j \in \ell_p. \quad (3)$$

Then $\|u_k\|_p = 1$ and $u_k \xrightarrow{\mathcal{D}} 0$ in ℓ_p .

Proof. The norm assertion is immediate. Let (P_{π_k}) be an arbitrary sequence in $\mathcal{D}$. The vector $P_{\pi_k} u_k$ equals $k^{-1/p}$ on a k -element set $S_k \subset \mathbb{N}$ and vanishes elsewhere.

Write $p' = p/(p-1)$. Fix $f \in \ell_{p'}$ and a finite set $F \subset \mathbb{N}$. By the triangle inequality and Hölder's inequality,

$$\begin{aligned} |\langle P_{\pi_k} u_k, f \rangle| &\leq k^{-1/p} \sum_{j \in S_k \cap F} |f_j| + k^{-1/p} \sum_{j \in S_k \setminus F} |f_j| \\ &\leq k^{-1/p} \|f \mathbf{1}_F\|_1 + k^{-1/p} |S_k|^{1/p} \|f \mathbf{1}_{F^c}\|_{p'} \\ &= k^{-1/p} \|f \mathbf{1}_F\|_1 + \|f \mathbf{1}_{F^c}\|_{p'}. \end{aligned} \quad (4)$$

For fixed F , the first term tends to zero. The second can be made arbitrarily small by taking F sufficiently large, independently of the sets S_k . Therefore $P_{\pi_k} u_k \rightarrow 0$ for every permutation sequence, proving the $\mathcal{D}$ -weak convergence. $\square$

4 Failure of persistence

Theorem 1. *For the couples and group in (2), the endpoint embedding $A_1 \hookrightarrow B_1$ is $\mathcal{D}$ -cocompact. Nevertheless, for every $\theta \in (0, 1)$, if $p = 1/\theta$, then neither*

$$[A_0, A_1]_\theta \hookrightarrow [B_0, B_1]_\theta \quad (5)$$

nor

$$(A_0, A_1)_{\theta, p} \hookrightarrow (B_0, B_1)_{\theta, p} \quad (6)$$

is $\mathcal{D}$ -cocompact.

Proof. Endpoint cocompactness is Lemma 1. The standard sequence-space interpolation identities are [2]

$$[\ell_\infty, \ell_1]_\theta = \ell_p, \quad (\ell_\infty, \ell_1)_{\theta, q} = \ell_{p, q}, \quad \frac{1}{p} = \theta. \quad (7)$$

For $q = p$, $\ell_{p, p} = \ell_p$ with an equivalent norm. Thus both (5) and (6) are identity embeddings of ℓ_p into itself, up to equivalent norms. Lemma 2 supplies a $\mathcal{D}$ -weakly null sequence whose norm is constantly one. Neither embedding is $\mathcal{D}$ -cocompact. $\square$

This is a negative answer to Remark 1.7. The same example defeats the complex assertion and, at an allowed fine index, the real assertion for every interpolation parameter.

5 Checks, comparison, and novelty status

Quantifiers. The crucial estimate (4) is uniform in the permutation sequence. It therefore proves the universal quantifier in the source's definition of $\mathcal{D}$ -weak convergence, not merely weak convergence along one selected sequence of permutations.

Equivalent norms. Replacing the real-method norm by the equivalent standard ℓ_p norm does not change the continuous dual, weak convergence, or which sequences converge to zero in norm. Hence it cannot change cocompactness of the identity.

Relation to the source hypotheses. All baseline endpoint and group assumptions hold. The construction need not admit the source’s covariant mollifier family: that is an additional condition whose proposed removal is exactly what the question asks about. Indeed, the source theorem and Theorem 1 together imply that no such family can satisfy all its stated conditions for this example.

Bounded novelty audit. The run indexes contained no duplicate. Exact and close searches found no published answer. OpenAlex listed twelve works citing the source; their titles and available abstracts concern applications rather than this abstract question. Tintarev’s 2013 survey [4] still says one would expect a general interpolation principle and refers only to the conditional source theorem. That survey uses the same flat vectors to illustrate remainders for the shift group, but does not combine them with the full permutation group and the Schur endpoint. Novelty confidence is moderate-to-high under this bounded audit, not an exhaustive priority claim.

Human-review recommendation. Prioritize review. The proof is short and answers both interpolation methods. Check the exponent convention in (7), the head–tail estimate, and priority beyond the searched citation graph.

References

- [1] M. Cwikel and K. Tintarev, *On interpolation of cocompact imbeddings*, Rev. Mat. Complut. **26** (2013), 33–55; arXiv:1008.4371v3.
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- [3] J. Diestel, *Sequences and Series in Banach Spaces*, Springer, 1984.
- [4] C. Tintarev, *Concentration analysis and cocompactness*, arXiv:1309.3431, 2013.